

FIT Lecture # 8

Engr. Naveera Sami



Computers were originally developed as standalone, single-user systems. Stand-alone computers can receive user's data, manipulate them and provide useful information for making decisions.

Here, the user uses his own data for his own decision making purposes. When the use of computers spread across government offices and business organisations , a number of issues were raised.

What if a user wants to share his computer generated information with other colleagues?

What if a user wants to explore the possibility of using certain information stored elsewhere?

These issues were addressed by the subsequent developments in software, hardware and communication technologies which have enabled the computers to communicate between each other and exchange information quickly and accurately and at any time.

The process of electronic transfer of information between two locations is known as data communication.

The five basic elements of data communication are:

Message It is the information to be communicated. It may be in the form of text, pictures, audio, video, or any combination of these.

Sender It is the device that creates and transmits the information.

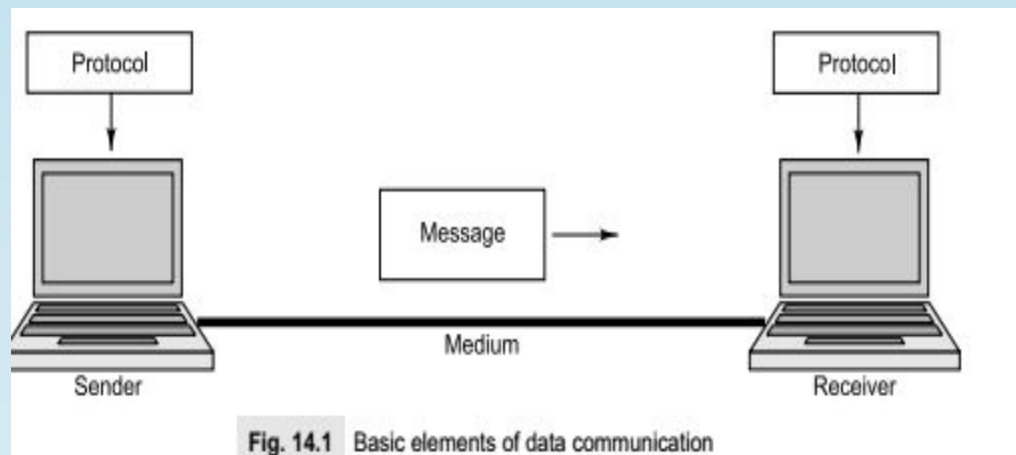
Receiver It is the device that receives the information.

Medium It is the communication channel through which the information travels from sender to receiver. It could be a physical wire or radio waves.

Protocol It represents a set of rules that governs the communication process between the sender and the receiver.

Figure 14.1 illustrates these five elements of data communication.

Figure illustrates these five elements of data communication. Data communication may occur in a simple point-to-point mode as shown in Fig. or in a multipoint mode where more than two computers are connected together in the form of a network. A network is a system of interconnected computers that can communicate with one another to share applications and data.



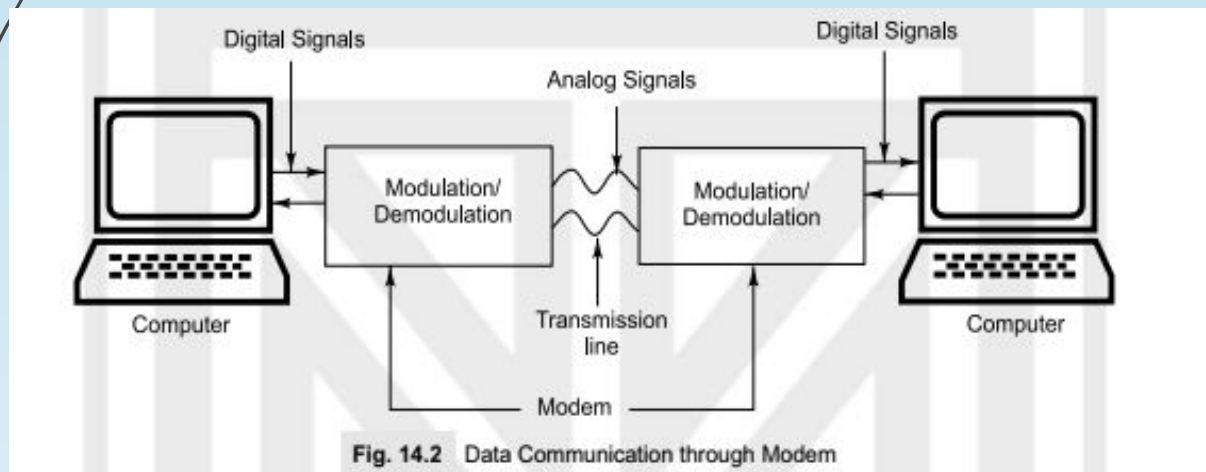
DATA COMMUNICATION USING MODEM

In data communication, data from one computer is transferred to another computer using various channels such as analog transmission channel and satellites. One of the analog transmission channels, which is most commonly used in data communication is the telephone lines. A computer that sends data to another computer is called the source terminal and the computer which receives the data from a sender computer is called the destination terminal.

The sender computer provides the data for transmission in the form of digital signals. These digital signals have to be first converted into analog signals for transmission through an analog transmission channel such as telephone lines. The process of converting digital signals into analog signals is called modulation.

The analog signals have to be again converted into digital signals when they reach the destination so that the receiver computer can understand the signals. The process of converting the analog signals again into digital signals is called demodulation. The modulation and demodulation processes are performed by a device called modem during data transfer between computers. A modem is basically an electronic device that converts the digital signals into analog signals. It also helps to convert analog signals into digital signals.

A modem must be present both at the source and destination for data transmission. This means a sender computer as well as a receiver computer must have a modem for data transmission. If a sender computer does not have a modem then it will not be able to send data to the receiver computer through an analog transmission channel. Similarly, if a receiver computer does not have a modem then it will not be able to receive the data sent by a sender computer through analog transmission channel. Figure shows the use of modem in data communication.



When using a modem in data communication, there are two factors that are of importance, modem speed and modem type. Modem speed refers to the time period that a modem takes to convert digital signals into analog signals or analog signals into digital signals. The speed of the modem is very important for data transmission between the sender computer and receiver computer. If a modem is able to convert digital signals into analog signals or analog signals into digital signals at a high speed then data will also be transmitted at high speed. A modem's speed is measured in bits per second (bps).

Modem type refers to the type of a modem. There are basically two types of modems: internal modem and external modem.

Internal modem is the modem that is present on the expansion board, which is contained in a computer. An external modem on the other hand is a modem that is connected externally to the COM port of the computer through a cable. Normally internal modems are considered better than external modems because these types of modems are present inside a computer and save desktop space.

Windows operating system provides built-in mechanism for setting up a connection using modem. We can either set up a direct connection or a dial-up connection using a modem. In Windows operating system, a new connection through a modem is established using the New Connection Wizard. Figure shows the welcome screen of the New Connection Wizard.

COMPUTER NETWORK

Computer network is a system of interconnected computers that enable the computers to communicate with each other and share their resources, data and applications. The physical location of each computer is tailored to personal and organizational needs. A network may include only personal computers or a mix of PCs, minis and mainframes spanning a particular geographical area. Computer networks that are commonly used today may be classified as follows:

COMPUTER NETWORK

Based on geographical area:

- o Local Area Networks (LANs)
- o Wide Area Networks (WANs)

Metropolitan Area Networks (MANs)

- o International Network (Internet)
- o Intranet

Based on how computer nodes are used:

- o Client Server Networks (CSNs)
- o Peer-to-peer Networks (PPNs)
- o Value-added Networks (VANs)

Local Area Network (LAN)

LAN is a group of computers, as shown in Fig., that are connected in a small area such as building, home, etc. Through this type of network, users can easily communicate with each other by sending and receiving messages. LAN is generally used for connecting two or more personal computers through some medium such as twisted pair, coaxial cable etc. Though the number of computers connected in a LAN is limited, the data is transferred at an extremely faster rate.

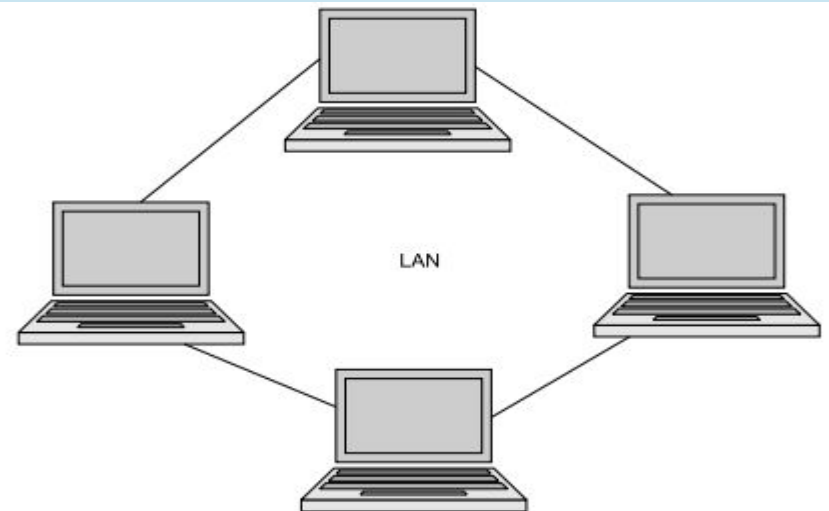


Fig. 14.4 A LAN

Wide Area Network (WAN)

WAN is a group of computers that are connected in a large area such as continent, country, etc.

WAN is generally used for connecting two or more LANs through some medium such as leased telephone lines, microwaves, etc. In WAN, data is transferred at slow rate. A typical WAN network is shown in Fig.

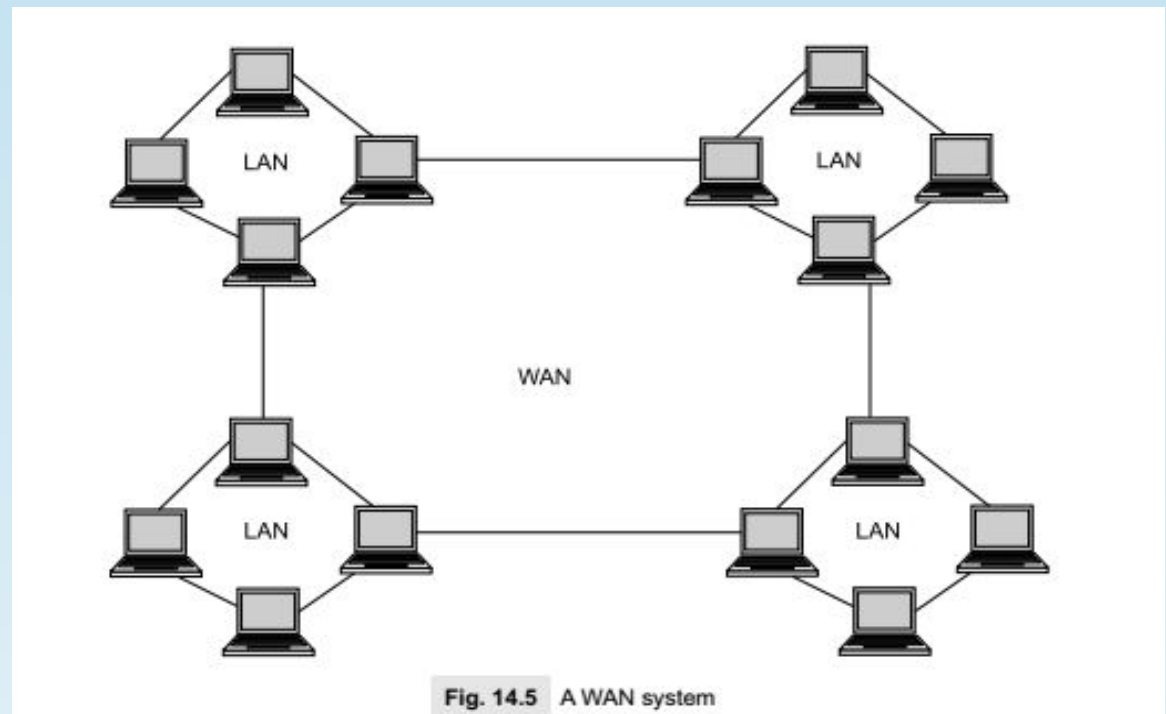
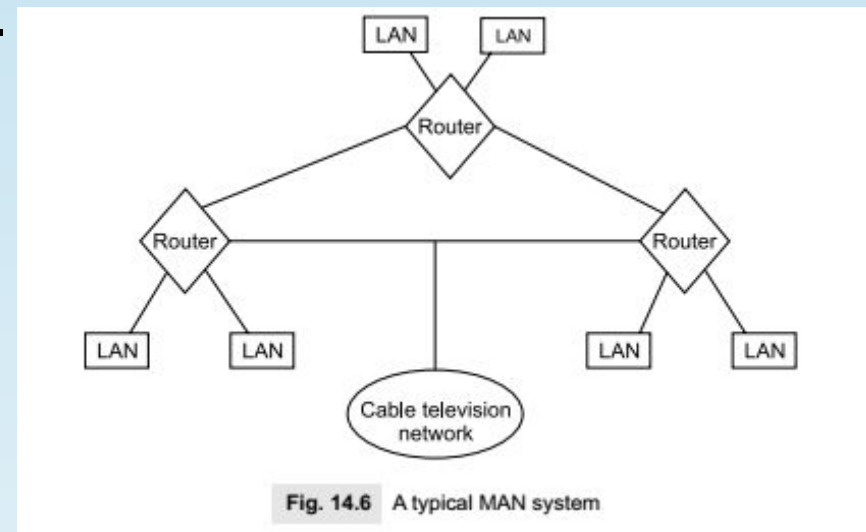


Fig. 14.5 A WAN system

Metropolitan Area Network (MAN)

MAN is a network of computers that covers a large area like a city. The size of the MAN generally lies between that of LAN and WAN, typically covering a distance of 5 Km to 50 Km. The geographical area covered by MAN is comparatively larger than LAN but smaller than WAN. MAN is generally owned by private organisations. MAN is generally connected with the help of optical fibres, copper wires etc. One of the most common example of MAN is cable television network within a city as shown in Fig A network device known as router is used to connect the LANs together. The router directs the information packets to the desired destination.



Thanks

